

LDMS USER GROUP MEETING

May 18, 2026

Agenda

- LDMSCON2026 News
- V4.5.2
- Q&A

LDMSCON2026 News

LDMSCON2026

 David Rubenstein Forum · University of Chicago · Chicago, IL

 June 9–11, 2026

 Sessions: 9:00 AM – 5:00 PM daily

- Check-in starts at 8:30 on June 9

 Special evening event: Wednesday, June 10

- If you plan bringing a plus one, please get a ticket on Eventbrite
<https://ldmscon2026.eventbrite.com/>

Registration

 In-Person Registration Still open — late fee applies

 Remote Registration Still open

Registration Link: <https://ldmscon2026.eventbrite.com/>

What to Expect

-  Presentations (30 min): Community results, deployments, and operational experience
-  Lightning Talks (15 min): Fast takes on new ideas, early work, and specific techniques
-  Focus Area Discussions (60–90 min): Structured community conversations on current challenges; active participation encouraged
-  Attendee-Led Discussions: Topics decided day-of; follow the conference energy and dig deeper
-  Open Interaction and Troubleshooting: Unstructured, free-time to get together in small groups

Program – Multi-Tenant Monitoring & Per-Job metrics

Per-Job & Multi-Tenant Monitoring

- Per-job data collection, PID tracking, and data access policy
- Security & authentication across container boundaries — working group report & community input
- User-driven monitoring deployments

Program – Core & Infrastructure

LDMS Core & Roadmap

- 2026 year in review · Road to LDMS 2027
- NERSC-sponsored enhancements
- Monitoring the monitoring infrastructure
- Low-latency feedback loops via LDMS message channel

Deployment & Configuration at Scale

- YAML and Ansible config management across a large HPC center
- Containerized development and deployment workflows
- Grafana containerization
- Advanced configuration best practices across heterogeneous platforms (LANL vs. SNL)
- Maestro for configuration & load balancing

Program – Intelligence & Hardware



Anomaly Detection & ML

- ML-based anomaly detection on contaminated production data
- Time series analysis for job fault prediction
- Generative AI for synthetic HPC log generation



Hardware-Level Telemetry

- Always-on MPI spinlock & load imbalance detection via PMU counters
- Memory bandwidth visibility via perfevent2 on Sapphire Rapids

Program – Tooling & Storage



Data Analysis & Tooling

- Portable job characterization pipelines on a single workstation
- PID data analysis at hyperscale
- Job scorecarding for post-run performance attribution
- AI-assisted deployment documentation



Storage (Focus Area Discussion)

- Benchmarking and comparing back-end storage technologies for monitoring and analysis

V4.5.2

V4.5.2 Released

✓ V4.5.2 has been tagged

📄 GitHub Release Page: <https://github.com/ovis-hpc/ldms/releases/tag/v4.5.2>

🔑 Highlights (covered in detail last meeting)

- RDMA now works on OmniPath and InfiniBand — backward compatible with v4.5.1
- `message_tag` is now the preferred keyword in `prdc_r_subscribe/prdc_r_unsubscribe` — `message_channel` remains a valid alias, no immediate config changes required
- Bug fixes across sampler and storage plugins, YAML parser, decomposition, LDMS transport and Messages, and build scripts

🙏 Thank you

- All contributors and bug reporters

🔭 What's next

- No known issues requiring a v4.5.3 at this time
- Focus shifts to new feature testing on the main branch

Q & A

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Thank you!